Progress Report

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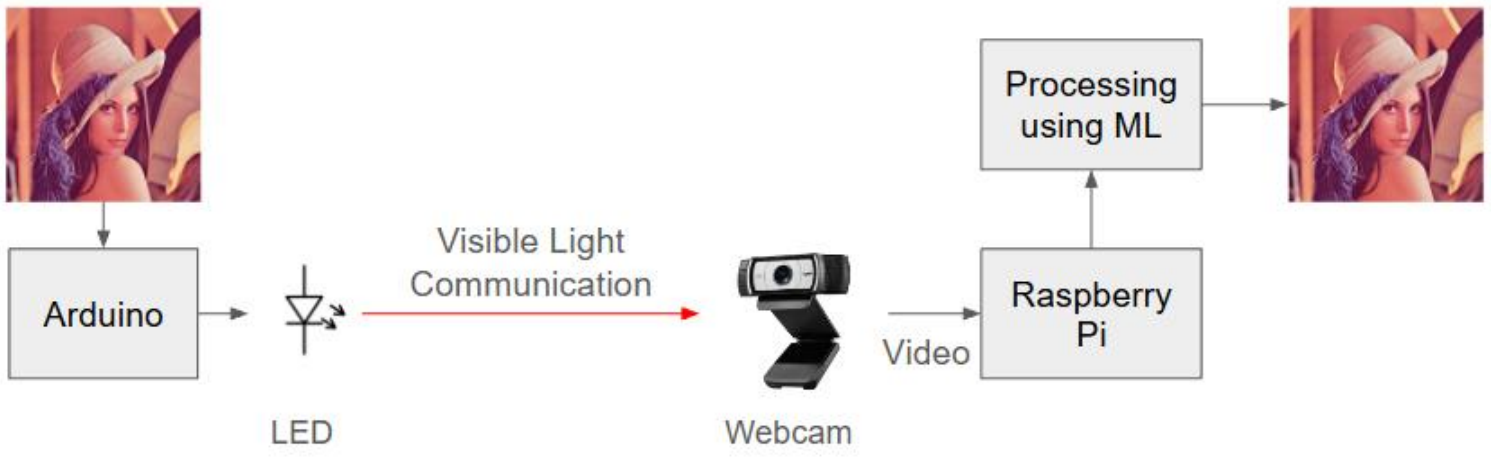
Overview of Research and Current Status

•**Objective:** To build a basic experimental system for Visible Light Communication (2-PAM/OOK) and confirm the principles from transmission to demodulation.
•**System Configuration:**
•**Tx:** PC sends bit sequence $¥rightarrow$ Arduino blinks LED (OOK modulation).
•**Channel:** LED blinking is recorded as video by a smartphone camera.
•**Rx:** demodulator.py extracts luminance signal from video and demodulates it.
•**Current Status:** The system is operational, but **Bit Error Rate (BER) is high and unstable** due to synchronization issues.



Overview of demodulator.py

- このコードは、動画から可視光通信（2-PAM/OOK）の信号を復調する受信側プログラムです。特に、「[2]On_Using_the_Raspberry_Pi_Camera_to_Receive_8-PAM_Signals_in_OCC.pdf」の手法（微分によるエッジ検出と相互相関による同期）を実装しています。

- This code is a receiver-side program that demodulates visible light communication (2-PAM/OOK) signals from a video. Specifically, it implements the method of "[2]On_Using_the_Raspberry_Pi_Camera_to_Receive_8-PAM_Signals_in_OCC.pdf" (synchronization using differentiation for edge detection and cross-correlation).

Main Processing Steps

- **Signal Extraction (extract_signal_from_video)**:
  - Reads the specified video file.
  - Asks the user to select the LED's light-emitting region (ROI) using mouse operations.
  - Calculates the average luminance of the selected region for all frames and extracts it as time-series data (raw signal).
- **Synchronization and Demodulation (demodulate_signal)**:
  - **Normalization**: Scales the luminance values of the signal to a range of 0 to 1.
  - **Synchronization (Rate and Phase Estimation)**:
    - Differentiates the signal to emphasize change points (edges).
    - Searches for the optimal symbol rate (number of frames per symbol) and the optimal reading start position (phase) around the "estimated frame rate" input by the user (±30%).
    - The search criterion is the timing when the average absolute value of the differential value is maximized (= edges are most concentrated).
  - **Sampling**: Acquires the luminance value near the center of each symbol based on the determined rate and phase.
  - **Bit Determination**: Determines '1' if the acquired luminance value is greater than 0.5, and '0' if it is smaller.
  - **Preamble Detection**: Searches for the preamble (101011) from the beginning of the bit sequence to identify the start position of the data. It also has a retry function for cases where the signal is inverted.
- **BER Evaluation (evaluate_ber)**:
  - Compares the demodulated bit sequence with the pre-prepared ground truth file (ground_truth.txt).
  - Calculates and displays the Bit Error Rate (BER).
- **Result Plotting (plot_results)**:
  - Plots and displays the waveform of the raw signal, detected symbol boundaries, sampling points, etc. This allows for visual confirmation of whether synchronization was performed correctly.

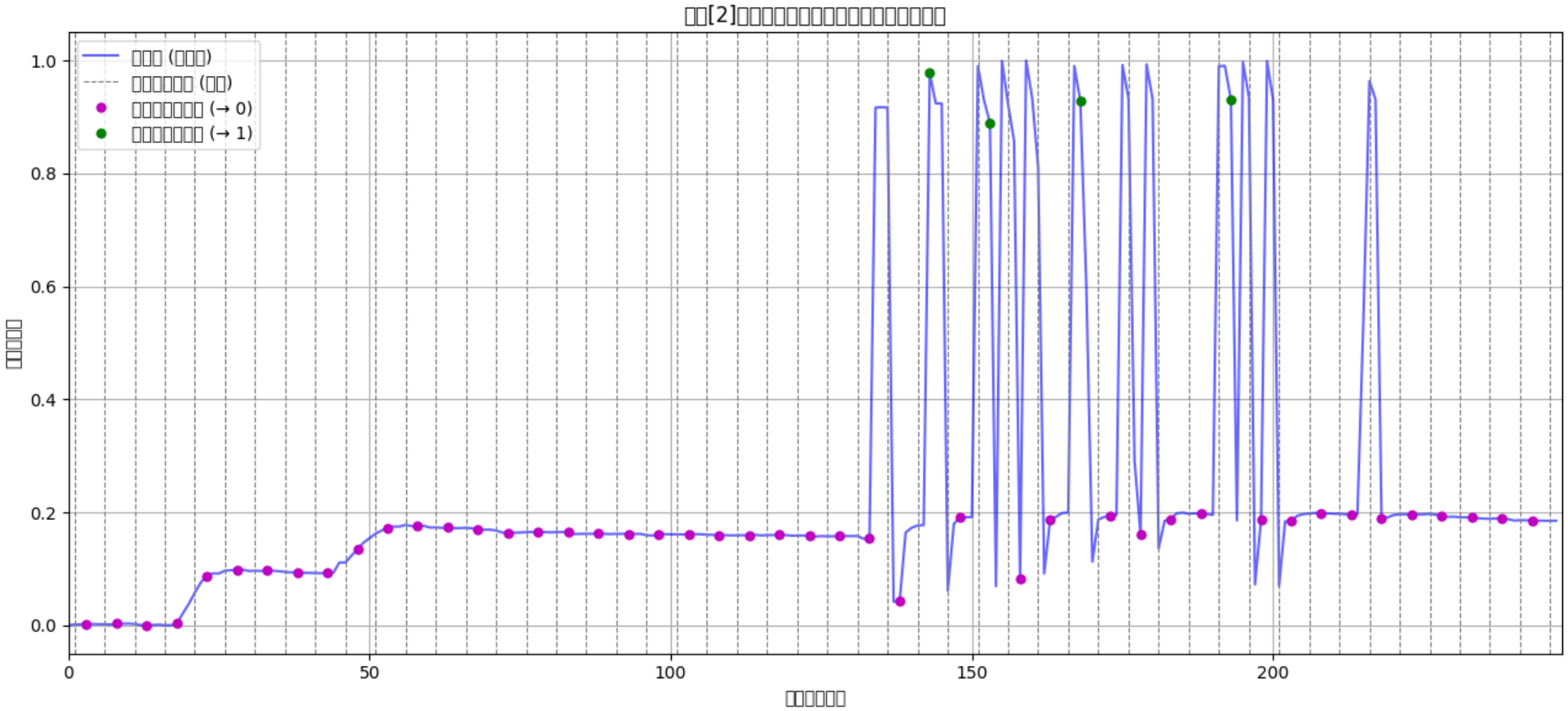
Summary

- **同期手法**: 「論文[2]の手法（微分してエッジの周期性を探る方法）で同期を試みています」
- **現状の問題**: 「送信レートとカメラのフレームレートが非整数の関係（例: 2.4フレーム/シンボル）になると、このコードの整数ベースの探索では同期がずれてしまい、BERが悪化します」
- **試した対策**: 「送信レートを調整して整数倍（3.0や4.0フレーム/シンボル）に近づけようとしましたが、PC側の送信タイミングの揺らぎで安定しませんでした」

- **Synchronization Method**: "I am attempting synchronization using the method from Paper [2] (a method of searching for periodicity of edges by differentiating)."
- **Current Problem**: "When the transmission rate and the camera's frame rate have a non-integer relationship (e.g., 2.4 frames/symbol), synchronization shifts with this integer-based search code, and BER deteriorates."
- **Attempted Countermeasures**: "I tried to adjust the transmission rate closer to an integer multiple (3.0 or 4.0 frames/symbol), but it did not stabilize due to fluctuations in transmission timing on the PC side."

Problem: Out of sync

True bit =
1 0 1 0 1 1 0 0 1 1 1 0 0 0 1 0



Problem: Out of sync

Current Issue (Synchronization Drift)

- **Problem:** "Synchronization drift" occurs where sampling positions gradually shift over time.
- **Evidence (wave.png):**
  - In the latter half of the graph, sampling points (● marks) shift from the center of the pulse to the edge.
  - Eventually, errors occur where "High" pulses are read at "Low" timing.

Root Cause Analysis

- **Root Cause:** Non-integer relationship between transmission rate and camera frame rate[8].
- Example: 1 symbol = 100ms, Camera = 33.3ms/frame
  -> 1 symbol ≈ **3.003 frames**.
- **Mechanism:** Since demodulator.py synchronizes in integer frames, a shift of 0.003 frames accumulates, eventually leading to a 1-frame shift and failure.

Proposed Countermeasures (For Discussion)

- **Option 1 (Software Improvement):** Implement resampling in demodulator.py to handle non-integer rates.
- **Option 2 (Hardware Improvement):** Improve Arduino program for more precise timing control, independent of PC delay.
- **Question:** Considering the current status, which approach should be prioritized?